

Dataset Analysis for Decision-Making

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the provided dataset (`sicor_municipal_contracts.csv`) and address any specific analytical question, I will first load the dataset and examine its structure to identify key characteristics. These characteristics include the number of rows and columns, column names, data types, missing values, and any notable patterns or distributions in the data.

Once the dataset is understood, I will address the specific analytical question using Python code to extract and process the relevant data. Below is the step-by-step approach:

Step 1: Load the Dataset and Inspect Its Structure

I will use Python's `pandas` library to load the dataset and inspect its structure.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values per Column:")
17 print(data.isnull().sum())
```

Step 2: Identify Key Characteristics

Based on the output from the above code, I will summarize:

- Number of Rows and Columns** : Total records and attributes.
- Column Names and Data Types** : What each column represents and its data type.
- Missing Values** : Any columns with missing or null values.
- Notable Patterns** : Initial observations about distributions, ranges, or relationships.

Step 3: Address the Analytical Question

Once you provide a specific analytical question, I will:

- Write Python code to extract and process the relevant data.
- Provide a clear and concise answer to the question based on the analysis.

Example Analytical Question

If the question were: "What is the total monetary value of contracts awarded in the dataset?", the solution would look like this:

Python Code:



```
python
1 # Calculate the total monetary value of contracts
2 total_contract_value = data['contract_value'].sum()
3
4 # Print the result
5 print(f"Total Monetary Value of Contracts: ${total_contract_value:,.2f}")
```

Explanation:

- The column `contract_value` (assumed to exist in the dataset) is summed up to calculate the total value of all contracts.
- The result is formatted to display as currency.

Next Steps

Please provide the specific analytical question you would like addressed using the dataset. I will then proceed with the detailed analysis and Python code tailored to your request.

How can the characteristics of a state's contracts influence the increase in the value allocated to industrialization? Define three options to address this issue and select the best one.



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To address the question of how the characteristics of a state's contracts can influence the increase in the value allocated to industrialization, we need to analyze the dataset (`sicor_municipal_contracts.csv`) for patterns and relationships. Below, I outline three potential analytical approaches to investigate this issue. After defining these options, I will select the best one based on feasibility and relevance to the dataset.

Step 1: Understanding the Dataset

Before proposing options, let's assume the dataset contains relevant information about contracts, such as:

- `state` : The state where the contract was awarded.
- `contract_value` : The monetary value of the contract.
- `sector` : The sector or industry the contract pertains to (e.g., industrialization, infrastructure, services).
- `year` : The year the contract was awarded.
- Other metadata like `contract_type` , `duration` , or `awarding_entity` .

If these assumptions hold, we can proceed with the analysis. If not, please clarify the dataset's structure.

Step 2: Three Analytical Options

Option 1: Correlation Analysis Between Contract Characteristics and Industrialization Spending

- **Approach** : Identify states that allocate significant resources to industrialization by filtering contracts in the "industrialization" sector. Then, analyze correlations between contract characteristics (e.g., total contract value, number of contracts, average contract size) and the growth in industrialization spending over time.
- **Key Metrics** :
 - Total contract value per state.
 - Number of contracts awarded per state.
 - Year-over-year growth in industrialization spending.
- **Outcome** : Determine whether larger or more frequent contracts correlate with increased industrialization spending.

Option 2: Sectoral Allocation Trends Across States

- **Approach** : Compare the proportion of contracts allocated to industrialization versus other sectors (e.g., infrastructure, services) across states. Analyze whether states with a higher proportion of industrialization contracts experience greater increases in industrialization spending.
- **Key Metrics** :
 - Percentage of total contract value allocated to industrialization.
 - Percentage of total contracts allocated to industrialization.
 - Growth rate of industrialization spending.



- **Outcome :** Identify whether prioritizing industrialization in contract allocation leads to higher industrialization spending.

Option 3: Time-Series Analysis of Industrialization Spending

- **Approach :** Conduct a time-series analysis of industrialization spending for each state. Examine how changes in contract characteristics (e.g., contract size, frequency, sector focus) precede or coincide with increases in industrialization spending.
- **Key Metrics :**
 - Year-over-year change in industrialization spending.
 - Lagged effects of contract characteristics on spending trends.
- **Outcome :** Assess whether specific contract characteristics predict future increases in industrialization spending.

Step 3: Selecting the Best Option

The best option depends on the dataset's structure and the availability of relevant variables. Assuming the dataset includes:

- A `sector` column indicating whether a contract pertains to industrialization.
- A `year` column for temporal analysis.
- A `contract_value` column for monetary analysis.

Selected Option: Option 2 (Sectoral Allocation Trends Across States)

Reasoning :

- This option is straightforward and directly addresses the relationship between contract allocation and industrialization spending.
- It leverages categorical data (sector) and numerical data (contract value) to provide actionable insights.
- It avoids the complexity of time-series modeling (as in Option 3) while still capturing meaningful trends.

Step 4: Implementation of Selected Option

Below is the Python code to implement Option 2:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Filter contracts related to industrialization
8 industrial_contracts = data[data['sector'] == 'industrialization']
9
10 # Calculate total contract value and count per state
11 state_summary = industrial_contracts.groupby('state').agg(
12     total_contract_value=('contract_value', 'sum'),
13     total_contracts=('contract_value', 'count')
14 ).reset_index()
15
16 # Merge with overall contract data to calculate proportions
17 overall_summary = data.groupby('state').agg(
18     total_overall_value=('contract_value', 'sum'),
19     total_overall_contracts=('contract_value', 'count')
20 ).reset_index()
21
22 state_analysis = pd.merge(state_summary, overall_summary, on='state')
23
24 # Calculate proportions
25 state_analysis['industrial_value_proportion'] = (
26     state_analysis['total_contract_value'] / state_analysis['total_overall_value']
27 )
28 state_analysis['industrial_contract_proportion'] = (
29     state_analysis['total_contracts'] / state_analysis['total_overall_contracts']
30 )
31
32 # Display results
33 print("State-wise Analysis of Industrialization Contract Allocation:")
34 print(state_analysis[['state', 'industrial_value_proportion', 'industrial_contract_proportion']])
```

Step 5: Interpretation of Results

1. **High Proportions of Industrialization Contracts :** States with a high proportion of industrialization contracts (both in



value and count) are likely prioritizing industrialization.

2. **Correlation with Spending Growth** : Cross-reference these proportions with external data on industrialization spending growth (if available) to identify states where contract allocation correlates with increased spending.
3. **Policy Implications** : States with low proportions of industrialization contracts may benefit from reallocating resources to this sector.

Conclusion

The selected option (Option 2) provides a clear and actionable analysis of how contract allocation influences industrialization spending. By focusing on sectoral proportions, we can identify states that prioritize industrialization and assess their outcomes. If additional details about the dataset or specific analytical requirements are available, I can refine the analysis further.

